Implementation Module Reference

Module	Primary responsibility
config/settings.py	Pydantic BaseSettings, environment overrides and storage paths
core/logger.py	Structured logging
core/ingestion/pdf_extractor.py	PyMuPDF document/pages/blocks extraction
core/ingestion/cleaner.py	Hyphenation, whitespace, recurring header/footer cleaning
core/chunking/hierarchical_chunker.py	Structural/section chunks
core/embeddings/embedder.py	Pretrained Sentence Transformers inference
core/vector_store/chroma_manager.py	Chroma persistent index, vector lookups
core/retrieval/retriever.py	Ranked chunks, top-k and optional threshold
core/rag/prompt_builder.py	Evidence delimitation and grounding instructions
core/llm/base.py	Provider-neutral response interface
core/llm/mock_provider.py	Deterministic mock response
core/llm/gemini_provider.py	Gemini inference
core/llm/factory.py	Provider selection
core/rag/citation_engine.py	Valid/invalid citation ID mapping
core/extraction/entity_extractor.py	SWC/Port/Interface candidates, validation, deduplication
storage/database.py	SQLite access/upserts
backend/coordinator.py	Ingest, query, entities orchestration
backend/main.py	FastAPI endpoints
frontend/app.py	UI upload, question and entity tables
core/evaluation/evaluator.py	Dataset-based proxy metrics
tests	Component/API/UI regression tests